Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(b)	(i)		eight —OH groups react 8 × 0.0700 mol of ethanoic anhydride needed = 0.560 mol (1) mass = 102 × 0.560 = 57.1 g minimum volume = $\frac{57.1}{1.08}$ = 52.9 cm^3 (1)	1	1		2	1	
		(ii)		award (1) each for any two of following percentage yield reaction rate / time needed need of a catalyst energy costs – must be qualified (and not related to heating)			2	2		
	(c)	(i)		red-brown precipitate/solid	1			1		1
		(ii)		aldehyde / CHO	1			1		1
				Question 10 total	**3**	**4**	**5**	**12**	**3**	**4**